Minimal Dominating Set Enumeration in Incomparability Graphs of Restricted Posets

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Abstract

A minimal dominating set of a graph G is a dominating set containing no proper dominating subset, and Dom-Enum asks to enumerate all minimal dominating sets of G. Motivated by the work of Bonamy, Defrain, Micek and Nourine [Discrete Math. 349 (2026), 114904] on incomparability graphs of bounded dimension posets, we study Dom-Enum for incomparability graphs of posets forbidding fixed standard examples. For $t \geq 1$, let S_t be the standard example with elements $a_1, \dots, a_t, b_1, \dots, b_t$ such that $a_i < b_j$ exactly when $i \neq j$.

We resolve two conjectures raised in the above work. First, we prove that, for every fixed integer t, Dom-Enum admits an output-polynomial time algorithm on incomparability graphs of S_t-free posets. Second, we prove that, for every fixed integer p, Dom-Enum admits an output-polynomial time algorithm on $2K_p$-free incomparability graphs, where $2K_p$ denotes the disjoint union of two cliques of size p. In addition, we obtain a partial result toward their broader conjecture on H-free incomparability graphs for fixed co-bipartite graphs H: we cover every fixed graph H that is an induced subgraph of the incomparability graph of S_q for some integer q.

Keywords: Minimal dominating sets; incomparability graphs; posets; algorithms
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1 Introduction

Enumeration problems ask for all objects of a prescribed type, rather than for one optimal object. Since the output may be exponentially larger than the input, polynomial time in the input size alone is not the appropriate measure of efficiency. The standard goal is output-polynomial time: the running time should be bounded by a polynomial in the size of the input and the size of the output.

In this paper we consider the enumeration of minimal dominating sets. A *dominating set* of a graph G is a set $D \subseteq V(G)$ such that every vertex of G belongs to D or has a neighbor in D. It is *minimal* if no proper subset of D is a dominating set. We write Dom-Enum for the problem of enumerating all minimal dominating sets of a given graph.

The problem Dom-Enum is one of the basic enumeration problems in graph algorithms, and its difficulty is reflected in its relation with hypergraph dualization. The minimal dominating sets of a

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graph are exactly the minimal transversals of its closed-neighborhood hypergraph. Conversely, Kanté et al. [12] proved that the general problem of enumerating all minimal transversals of a hypergraph reduces, in an output-polynomial sense, to Dom-Enum. Thus an output-polynomial algorithm for Dom-Enum on all graphs would resolve the long-standing open problem of whether all minimal transversals of a hypergraph can be enumerated in output-polynomial time. The best known general algorithms for hypergraph dualization are still quasi-polynomial, following the work of Fredman and Khachiyan [7]; see also the survey of Eiter et al. [5].

Several positive results are known on restricted graph classes. For instance, Dom-Enum is output-polynomial on split graphs and on P_6-free chordal graphs [12], on chordal bipartite graphs [9], and on K_t-free graphs for every fixed t [1]. More generally, width parameters provide another source of tractability: minimal dominating sets can be enumerated in output-polynomial time in graphs given together with a linear ordering of bounded linear MIM-width [8]. General exponential-time bounds and enumeration algorithms were obtained by Fomin et al. [6], and combinatorial bounds for several graph classes were studied by Couturier et al. [3].

We focus on incomparability graphs of posets. For a poset P, the *incomparability graph* $\mathrm{Inc}(P)$ has the elements of P as vertices, with two vertices adjacent exactly when they are incomparable in P. Thus incomparability graphs are precisely *co-comparability graphs*. This class forms a natural meeting point between order theory and graph algorithms: comparability graphs can be recognized and transitively oriented in polynomial time [10], while incomparability graphs encode the antichain structure of partial orders.

Bonamy, Defrain, Micek and Nourine [2] studied Dom-Enum on comparability and incomparability graphs arising from posets of bounded dimension. Among other results, they proved that Dom-Enum is output-polynomial on incomparability graphs of bounded dimension posets, when a bounded-dimensional representation is given. Their algorithm uses the geometric representation of such graphs due to Golumbic et al. [11]. They then asked whether the bounded dimension assumption can be relaxed.

For $t \geq 1$, the *standard example* S_t is the poset with elements $A = \{a_1, \ldots, a_t\}$ and $B = \{b_1, \ldots, b_t\}$ such that $a_i < b_j$ if and only if $i \neq j$. Thus $a_i \parallel b_i$ for every i, while all non-diagonal pairs between A and B are comparable. Standard examples are classical objects in the dimension theory of posets, going back to Dushnik and Miller [4]; see also Trotter [13]. Since large standard examples force large dimension, forbidding a fixed S_t is a natural way to go beyond bounded dimension.

Bonamy et al. formulated the following conjecture.

Conjecture 1.1 ([2, Conjecture 7.1])**.** *For every fixed integer t, Dom-Enum admits an output-polynomial time algorithm on incomparability graphs of S_t-free posets.*

Our first main result confirms this conjecture.

Theorem 1.2. *For every fixed integer $t \geq 1$, there is an output-polynomial time algorithm for Dom-Enum in incomparability graphs of S_t-free posets.*

They also proposed a related conjecture for incomparability graphs excluding two disjoint cliques.

Conjecture 1.3 ([2, Conjecture 7.2]). *For every fixed integer p, Dom-Enum admits an output-polynomial time algorithm on $2K_p$-free incomparability graphs, where $2K_p$ denotes the disjoint union of two cliques of size p.*

We also confirm this conjecture.

Theorem 1.4. *For every fixed integer $p \geq 1$, there is an output-polynomial time algorithm for Dom-Enum in $2K_p$-free incomparability graphs.*

Bonamy et al. further proposed a broader conjecture covering arbitrary fixed co-bipartite forbidden graphs, that is, graphs whose complements are bipartite.

Conjecture 1.5 ([2, Conjecture 7.3]). *For every fixed co-bipartite graph H, there is an output-polynomial time algorithm for Dom-Enum in H-free incomparability graphs.*

Our method also gives a partial answer of Conjecture 1.5. Recall that $M_q = \mathrm{Inc}(S_q)$.

Theorem 1.6. *Let H be a fixed graph. Suppose that H is an induced subgraph of M_q for some integer q. Then there is an output-polynomial time algorithm for Dom-Enum in H-free incomparability graphs.*

The proof is based on a structural observation connecting standard examples with induced matchings. Let P be a poset, let $G = \mathrm{Inc}(P)$, and let σ be a linear extension of P. If G contains an induced matching of size t across a cut of σ, then P contains an induced copy of S_t. Hence, for an S_t-free poset, every linear extension gives an ordering of the incomparability graph with linear MIM-width at most $t-1$. Thus Theorem 1.2 follows from the enumeration theorem for graphs of bounded linear MIM-width [8].

To obtain graph-level consequences, we use the graph $M_t = \mathrm{Inc}(S_t)$. We prove that, for co-comparability graphs, being the incomparability graph of an S_t-free poset is equivalent to being M_t-free. This yields Theorem 1.4, since M_{2p} contains an induced copy of $2K_p$. It also yields Theorem 1.6, since forbidding a graph H with $H \subseteq_{\mathrm{ind}} M_q$ forces the input graph to be M_q-free.

The paper is organized as follows. In Section 2 we recall the necessary definitions and the enumeration result for graphs of bounded linear MIM-width. In Section 3 we prove the main theorems, including the S_t-free case, the $2K_p$-free case, and the co-bipartite H-free subcase. In Section 4 we give concluding remarks.

2 Preliminaries

All graphs considered in this paper are finite, simple and undirected, and all posets are finite.

2.1 Graphs and domination

For a graph G, we write $V(G)$ and $E(G)$ for its vertex set and edge set. For a vertex $v \in V(G)$, let $N_G(v)$ denote the set of neighbors of v, and put $N_G[v] = N_G(v) \cup \{v\}$. For $D \subseteq V(G)$, write

$N_G[D] = \bigcup_{v \in D} N_G[v]$. We denote by $\mathcal{D}(G)$ the family of all minimal dominating sets of G. The problem Dom-Enum asks, given a graph G, to enumerate all sets in $\mathcal{D}(G)$. An algorithm for Dom-Enum runs in *output-polynomial time* on a graph class $\mathcal{C}$ if, for every input $G \in \mathcal{C}$, its running time is bounded by a polynomial in $|V(G)| + |\mathcal{D}(G)|$.

For a graph H, we say that a graph G is H-free if G contains no induced subgraph isomorphic to H. We write $2K_p$ for the disjoint union of two cliques of size p. A graph is *co-bipartite* if its complement is bipartite; equivalently, its vertex set can be partitioned into two cliques.

2.2 Posets and incomparability graphs

A *poset* is a pair $P = (V, \leq)$, where $\leq$ is a reflexive, antisymmetric and transitive relation on V. For distinct elements $x, y \in V$, we write $x < y$ if $x \leq y$. Two distinct elements $x, y \in V$ are *comparable* if $x \leq y$ or $y \leq x$, and *incomparable*, denoted by $x \parallel y$, otherwise.

For $X \subseteq V$, the induced subposet $P[X]$ is obtained by restricting $\leq$ to X. A poset P is Q-free if it contains no induced subposet isomorphic to Q.

The *incomparability graph* of P, denoted by $\mathrm{Inc}(P)$, is the graph with vertex set V in which two distinct vertices are adjacent exactly when they are incomparable in P; that is, $xy \in E(\mathrm{Inc}(P))$ if and only if $x \parallel y$. The *comparability graph* of P is the complement of $\mathrm{Inc}(P)$; its edges join the distinct comparable pairs of P.

A graph is a *co-comparability graph* if its complement is a comparability graph. Equivalently, G is a co-comparability graph if $G = \mathrm{Inc}(P)$ for some poset P. A transitive orientation of a graph F is an orientation of its edges such that, whenever xy and yz are oriented as $x \to y$ and $y \to z$, and $xz \in E(F)$, the edge xz is oriented as $x \to z$. Comparability graphs are exactly the graphs admitting a transitive orientation. We shall use the standard fact that comparability graphs can be recognized in polynomial time and, when the input graph is a comparability graph, a transitive orientation can be found in polynomial time [10].

A *linear extension* of a poset $P = (V, \leq)$ is a linear order σ of V such that whenever $x < y$ in P, the element x appears before y in σ.

2.3 The standard example

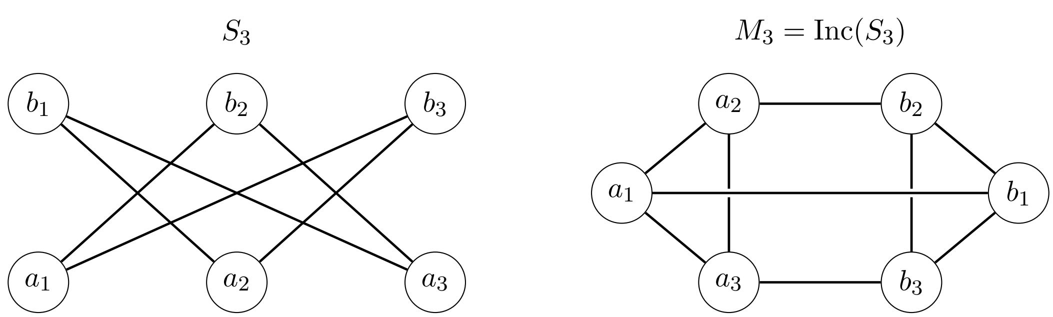

Figure 1: The standard example S_3 and its incomparability graph $M_3 = \mathrm{Inc}(S_3)$.

For $t \geq 1$, the *standard example* S_t is the poset with elements $A = a_1, \dots, a_t$ and $B = b_1, \dots, b_t$, where A and B are antichains, and $a_i < b_j$ if and only if $i \neq j$. Thus $a_i \parallel b_i$ for every i. A poset is S_t-free if it contains no induced subposet isomorphic to S_t.

Let $M_t = \mathrm{Inc}(S_t)$. Then M_t consists of two t-cliques $A = a_1, \dots, a_t$ and $B = b_1, \dots, b_t$, together with the matching edges $a_i b_i$, $i = 1, \dots, t$, and no other edges between A and B. Figure 1 illustrates the case $t = 3$.

2.4 Linear MIM-width

Let G be a graph and let $\sigma = (v_1, \dots, v_n)$ be a linear ordering of $V(G)$. For $1 \leq i < n$, put $L_i = \{v_1, \dots, v_i\}$ and $R_i = V(G) \setminus L_i$. Let $G[L_i, R_i]$ denote the bipartite graph formed by the edges of G with one endpoint in L_i and the other in R_i.

An *induced matching* in $G[L_i, R_i]$ is a set of pairwise disjoint edges $x_1y_1, \dots, x_ky_k$, with $x_j \in L_i$ and $y_j \in R_i$, such that there are no other edges of G between $\{x_1, \dots, x_k\}$ and $\{y_1, \dots, y_k\}$. Edges inside L_i and inside R_i are irrelevant for this definition. Let $\mathrm{mim}_G(L_i, R_i)$ be the maximum size of an induced matching in $G[L_i, R_i]$. The *linear MIM-width* of G with respect to σ is $\mathrm{lmim}_\sigma(G) = \max_{1 \leq i < n} \mathrm{mim}_G(L_i, R_i)$. We shall use the following theorem.

Theorem 2.1 ([8, Theorem 4])**.** *For every fixed integer k, there is an output-polynomial time algorithm for enumerating all minimal dominating sets in graphs given together with a linear ordering of linear MIM-width at most k.*

Finally, for every fixed graph H, testing whether an input graph contains an induced copy of H can be done in polynomial time by brute force.

3 Proofs of the main theorems

In this section we prove the two main theorems. The proof of the S_t-free case rests on the relation between standard examples and induced matchings across cuts of a linear extension. The $2K_p$-free case is then derived from this result.

3.1 The S_t-free case

The first lemma is the key structural observation. It shows that a large induced matching across a cut of a linear extension necessarily comes from a standard example.

Lemma 3.1. *Let $P = (V, \leq)$ be a poset, let $G = \mathrm{Inc}(P)$, and let σ be a linear extension of P. Suppose that, for some cut (L, R) of σ, the bipartite graph $G[L, R]$ contains an induced matching of size t, say $a_1b_1, \dots, a_tb_t$, where $a_i \in L$ and $b_i \in R$ for every i. Then $P[\{a_1, \dots, a_t, b_1, \dots, b_t\}]$ is isomorphic to S_t.*

Proof. Since $a_ib_i \in E(G)$, we have $a_i \parallel b_i$ for every i. Now fix $i \neq j$. As $a_1b_1, \dots, a_tb_t$ is an induced matching in $G[L, R]$, the pair a_ib_j is not an edge of $G[L, R]$. Hence a_i and b_j are comparable in P.

Since every vertex of L precedes every vertex of R in σ, the relation $b_j < a_i$ would contradict that σ is a linear extension. Hence $b_j < a_i$ is impossible, and therefore $a_i < b_j$.

It remains to show that the a_i's and the b_i's are antichains. Suppose that a_i and a_j are comparable for some $i \neq j$. If $a_i < a_j$, then $a_j < b_i$, and hence $a_i < a_j < b_i$, contradicting $a_i \parallel b_i$. The case $a_j < a_i$ is symmetric. Thus $a_i \parallel a_j$ for all $i \neq j$. The same argument, with the roles of A and B reversed, gives $b_i \parallel b_j$ for all $i \neq j$.

Thus the selected vertices form two antichains, the pairs a_i, b_i are incomparable, and $a_i < b_j$ whenever $i \neq j$. Hence they induce a copy of S_t. □

We next show that $M_t = \mathrm{Inc}(S_t)$ is a faithful graph obstruction.

Lemma 3.2. *Let P be a poset. If $\mathrm{Inc}(P)$ contains an induced copy of M_t, then P contains an induced copy of S_t.*

Proof. Let the induced copy of M_t have vertex set $A \cup B$, where $A = \{a_1, \ldots, a_t\}$ and $B = \{b_1, \ldots, b_t\}$ are the two cliques of M_t, and the only edges between A and B are $a_i b_i$, $i = 1, \ldots, t$. Since A and B are cliques in $\mathrm{Inc}(P)$, both A and B are antichains in P. Moreover, $a_i \parallel b_i$ for every i, while a_i and b_j are comparable whenever $i \neq j$.

For $t = 1$ the claim is immediate. For $t = 2$, the two comparable pairs are a_1, b_2 and a_2, b_1. If they have the same direction, then A and B are the two sides of an induced S_2, possibly after swapping the sides. If they have opposite directions, then $\{a_1, b_1\}$ and $\{a_2, b_2\}$ are the two sides of an induced S_2, again possibly after relabelling.

Assume now that $t \geq 3$. For each fixed i, all comparisons between a_i and the elements of $B \setminus \{b_i\}$ have the same direction. Indeed, if $a_i < b_j$ and $b_k < a_i$ for distinct $j, k \neq i$, then $b_k < a_i < b_j$, contradicting that B is an antichain. Hence, for each i, either $a_i < b_j$ for all $j \neq i$, or $b_j < a_i$ for all $j \neq i$.

These directions are globally uniform. Otherwise, for some distinct i, k we would have $a_i < b_j$ for all $j \neq i$ and $b_j < a_k$ for all $j \neq k$. Since $t \geq 3$, choose j distinct from both i and k. Then $a_i < b_j < a_k$, contradicting that A is an antichain. Therefore either $a_i < b_j$ for all $i \neq j$, or $b_j < a_i$ for all $i \neq j$. In the first case the selected vertices induce S_t, and in the second case they induce the dual of S_t, which is isomorphic to S_t after swapping the two sides. □

This gives the following graph-level characterization used in the algorithm.

Corollary 3.3. *Let G be a co-comparability graph. Then G is the incomparability graph of an S_t-free poset if and only if G is M_t-free.*

Proof. If $G = \mathrm{Inc}(P)$ for some S_t-free poset P and G contained an induced M_t, then Lemma 3.2 would give an induced S_t in P, a contradiction. Hence G is M_t-free.

Conversely, suppose that G is a co-comparability graph and is M_t-free. Choose a poset P such that $G = \mathrm{Inc}(P)$. If P contained an induced copy of S_t, then G would contain $\mathrm{Inc}(S_t) = M_t$, contrary to the assumption. Thus P is S_t-free. □

We are now in a position to prove the first main theorem.

Proof of Theorem 1.2. Fix t. Given a graph G, we first test whether $\overline{G}$ is a comparability graph and whether G is M_t-free. Since t is fixed, both tests can be performed in polynomial time. If one of the tests fails, we reject the input. For graphs in the class under consideration, both tests succeed.

Compute a transitive orientation of $\overline{G}$. This orientation defines a poset Q such that $G = \mathrm{Inc}(Q)$. Let σ be a linear extension of Q. We claim that $\mathrm{lmim}_\sigma(G) \le t-1$. Otherwise, for some cut (L, R) of σ, the bipartite graph $G[L, R]$ contains an induced matching of size t. By Lemma 3.1, the poset Q contains an induced copy of S_t, and hence $G = \mathrm{Inc}(Q)$ contains an induced copy of $M_t = \mathrm{Inc}(S_t)$, contradicting that G is M_t-free.

Thus σ is an ordering of G of linear MIM-width at most $t-1$. Since t is fixed, Theorem 2.1 enumerates all minimal dominating sets of G in output-polynomial time. □

3.2 The $2K_p$-free case

For the second theorem, we need the following elementary containment.

Lemma 3.4. *The graph M_{2p} contains an induced copy of $2K_p$.*

Proof. Let $A = \{a_1, \dots, a_{2p}\}$ and $B = \{b_1, \dots, b_{2p}\}$ be the two cliques of M_{2p}, with matching edges a_ib_i, $i = 1, \dots, 2p$. Put $A' = \{a_1, \dots, a_p\}$ and $B' = \{b_{p+1}, \dots, b_{2p}\}$. Then A' and B' are cliques of size p, and there is no edge between them, since the index sets are disjoint. Hence $M_{2p}[A' \cup B'] \cong 2K_p$. □

We now prove the second main theorem.

Proof of Theorem 1.4. Fix p, and let G be a $2K_p$-free incomparability graph. Then G is co-comparability. Moreover, G is M_{2p}-free, since otherwise Lemma 3.4 would give an induced copy of $2K_p$ in G. By Corollary 3.3, G is the incomparability graph of an S_{2p}-free poset. Applying Theorem 1.2 with $t = 2p$ gives an output-polynomial time algorithm for enumerating all minimal dominating sets of G. □

3.3 The co-bipartite H-free subcase

We now prove the subcase of Conjecture 1.5 stated in Theorem 1.6.

Proof of Theorem 1.6. Let G be an H-free incomparability graph. Since H is an induced subgraph of M_q, the graph G is M_q-free. Indeed, if G contained an induced copy of M_q, then this copy would also contain an induced copy of H, contrary to the assumption that G is H-free. By Corollary 3.3, the graph G is the incomparability graph of an S_q-free poset. The result follows from Theorem 1.2. □

The next corollary gives a concrete family of forbidden co-bipartite graphs covered by Theorem 1.6.

Corollary 3.5. *Let H be a fixed graph admitting a partition $V(H) = X \cup Y$ such that X and Y are cliques, and the edges between X and Y form a matching. Then there is an output-polynomial time algorithm for* DOM-ENUM *in H-free incomparability graphs.*

Proof. It is enough to show that H is an induced subgraph of M_q for some q. We may take $q = |V(H)|$. Let the two cliques of M_q be $A = \{a_1, \ldots, a_q\}$ and $B = \{b_1, \ldots, b_q\}$, with cross-edges exactly $a_i b_i$, $i = 1, \ldots, q$.

For each matching edge xy with $x \in X$ and $y \in Y$, assign one common index to x and y. Then assign pairwise distinct fresh indices to all remaining vertices, choosing these indices different from all indices already used by the matching edges. Map each vertex of X to the corresponding vertex of A, and each vertex of Y to the corresponding vertex of B.

This gives an injective map from $V(H)$ into $V(M_q)$. Edges inside X and inside Y are realized because A and B are cliques. For $x \in X$ and $y \in Y$, their images are adjacent in M_q if and only if they have the same index, which happens exactly when xy is one of the prescribed matching edges between X and Y. Hence H is an induced subgraph of M_q. The result follows from Theorem 1.6. □

Remark 3.6. The preceding argument does not settle the full co-bipartite H-free conjecture. It applies only to graphs H that occur as induced subgraphs of some M_q. Not every co-bipartite graph has this property. For example, $K_4 - e$ is co-bipartite, but it is not an induced subgraph of any M_q. Indeed, any four vertices of M_q induce either K_4, if they lie in one clique, or a graph with at most four edges if they meet both cliques, since the cross-edges between the two cliques form a matching. Since $K_4 - e$ has five edges, it cannot occur as an induced subgraph of M_q.

4 Concluding remarks

We proved two conjectures of Bonamy, Defrain, Micek and Nourine on the enumeration of minimal dominating sets in incomparability graphs. First, for every fixed integer t, DOM-ENUM is output-polynomial on incomparability graphs of S_t-free posets. Second, for every fixed integer p, DOM-ENUM is output-polynomial on $2K_p$-free incomparability graphs.

The main structural point is that standard examples in posets control induced matchings across cuts of a linear extension. After translating the order-theoretic obstruction S_t into the graph obstruction $M_t = \mathrm{Inc}(S_t)$, this gives a bounded linear MIM-width ordering for the relevant incomparability graphs.

We also proved a subcase of the broader co-bipartite H-free conjecture, namely the case where H is an induced subgraph of M_q for some q. Equivalently, this is precisely the case where H admits a partition into two cliques such that the cross-edges form a matching.

The full Conjecture 1.5 remains open. A natural case beyond the present method is the following.

Problem 4.1. *Is* DOM-ENUM *output-polynomial on $(K_4 - e)$-free incomparability graphs?*

References

[1] M. Bonamy, O. Defrain, M. Heinrich, J.-F. Raymond, and M. Pilipczuk, Enumerating minimal dominating sets in K_t-free graphs and variants, *ACM Trans. Algorithms* 16 (2020), Article 39, 1–23. 2

[2] M. Bonamy, O. Defrain, P. Micek, and L. Nourine, Enumerating minimal dominating sets in the (in)comparability graphs of bounded dimension posets, *Discrete Math.* 349 (2026), 114904. 2, 3

[3] J.-F. Couturier, P. Heggernes, P. van 't Hof, and D. Kratsch, Minimal dominating sets in graph classes: Combinatorial bounds and enumeration, *Theoret. Comput. Sci.* 487 (2013), 82–94. 2

[4] B. Dushnik and E. W. Miller, Partially ordered sets, *Amer. J. Math.* 63 (1941), 600–610. 2

[5] T. Eiter, K. Makino, and G. Gottlob, Computational aspects of monotone dualization: A brief survey, *Discrete Appl. Math.* 156 (2008), 2035–2049. 2

[6] F. V. Fomin, F. Grandoni, A. V. Pyatkin, and A. A. Stepanov, Combinatorial bounds via measure and conquer: Bounding minimal dominating sets and applications, *ACM Trans. Algorithms* 5 (2008), Article 9, 1–17. 2

[7] M. L. Fredman and L. Khachiyan, On the complexity of dualization of monotone disjunctive normal forms, *J. Algorithms* 21 (1996), 618–628. 2

[8] P. A. Golovach, P. Heggernes, M. M. Kanté, D. Kratsch, S. H. Sæther, and Y. Villanger, Output-polynomial enumeration on graphs of bounded (local) linear MIM-width, *Algorithmica* 80 (2018), 714–741. 2, 3, 5

[9] P. A. Golovach, P. Heggernes, M. M. Kanté, D. Kratsch, and Y. Villanger, Enumerating minimal dominating sets in chordal bipartite graphs, *Discrete Appl. Math.* 199 (2016), 30–36. 2

[10] M. C. Golumbic, *Algorithmic Graph Theory and Perfect Graphs*, 2nd ed., Ann. Discrete Math. 57, Elsevier, Amsterdam, 2004. 2, 4

[11] M. C. Golumbic, D. Rotem, and J. Urrutia, Comparability graphs and intersection graphs, *Discrete Math.* 43 (1983), 37–46. 2

[12] M. M. Kanté, V. Limouzy, A. Mary, and L. Nourine, On the enumeration of minimal dominating sets and related notions, *SIAM J. Discrete Math.* 28 (2014), 1916–1929. 2

[13] W. T. Trotter, *Combinatorics and Partially Ordered Sets: Dimension Theory*, Johns Hopkins Univ. Press, Baltimore, 1992. 2